

In-class Exercise 6

● Graded

Student

Jinku Jung

Total Points

4 / 4 pts

Autograder Score

4.0 / 4.0

Passed Tests

exception thrown (2/2)

add bad artist (2/2)

Question 2

Autograder adjustment

0 / 0 pts

✓ + 0 pts No Adjustment needed

+ 0 pts Adjustment needed

Autograder Results

exception thrown (2/2)

Tested play addCD for BadMusicException
BadMusicException correctly thrown

add bad artist (2/2)

Tested addBad with input: Cinderella

Submitted Files

```
1  /* Name: Jinku Jung
2   * UNI: dj2598
3   * Assignment: In-class Exercise #6
4   * Date: 4/8/2021
5   */
6
7  /*******
8   // Tunes.java    Author: Lewis/Loftus (modified by cannon)
9   //
10  // Demonstrates the use of an array of objects.
11  /*******
12
13  public class Tunes
14  {
15      //-----
16      // Creates a CDCollection object and adds some CDs to it. Prints
17      // reports on the status of the collection.
18      //-----
19      public static void main (String[] args)
20      {
21          CDCollection music = new CDCollection ();
22
23          //Bad artists are added to the "badArtists" blacklist here:
24
25          music.addBad("Nickelback");
26          music.addBad("Robert Palmer");
27
28          //Here I use a try ... catch statement to throw the exception
29          //handler, "BadMusicException," immediately after a bad artist
30          //on the blacklist is found in the containing commands.
31
32          try {
33              music.addCD ("Storm Front", "Billy Joel", 14.95, 10);
34              music.addCD ("Discipline", "King Crimson", 19.99, 26);
35              music.addCD ("Greatest Hits", "Jimmy Buffet", 15.95, 13);
36              music.addCD ("Trite Garbage ", "Nickelback", 15.00, 5);
37              music.addCD ("Come On Over", "Shania Twain", 1.95, 16);
38          } catch (BadMusicException e)
39          {
40              //Simply outputs the error message called from the
41              //BadMusicException class.
42
43              System.out.println(e.getMessage());
44          }
45
46          //Output, up until an exception error is thrown, is still
47          //displayed for the user to enjoy.
48
49          System.out.println(music);
```

```
50
51
52 }
53 }
54
```

▼ dj2598_ex6/BadMusicException.java

 Download

```
1
2  /* Name: Jinku Jung
3   * UNI: dj2598
4   * Assignment: In-class Exercise #6
5   * Date: 4/8/2021
6   */
7
8  /* *****
9   *
10  * This exception class should be *
11  * instantiated and thrown when *
12  * someone attempts to add bad *
13  * to a CDCollection object. *
14  * *****/
15
16 public class BadMusicException extends IllegalArgumentException{
17
18     public BadMusicException(){
19         super();
20     }
21
22     public BadMusicException(String message){
23         super(message);
24     }
25
26 }
27
```

```
1  /* Name: Jinku Jung
2   * UNI: dj2598
3   * Assignment: In-class Exercise #6
4   * Date: 4/8/2021
5   */
6
7  //*****
8  // CD.java    Author: Lewis/Loftus  (modified by cannon)
9  //
10 // Represents a compact disc.
11 //*****
12
13 import java.text.NumberFormat;
14 import java.util.Locale;
15
16 public class CD implements Comparable<CD>
17 {
18     private String title, artist;
19     private double cost;
20     private int tracks;
21
22     //-----
23     // Creates a new CD with the specified information.
24     //-----
25     public CD (String name, String singer, double price, int numTracks)
26     {
27         title = name;
28         artist = singer;
29         cost = price;
30         tracks = numTracks;
31     }
32
33     //-----
34     // Returns a string description of this CD.
35     //-----
36
37     public String toString()
38     {
39         NumberFormat fmt = NumberFormat.getCurrencyInstance(Locale.US);
40
41         String description;
42
43         description = fmt.format(cost) + "\t" + tracks + "\t";
44         description += title + "\t" + artist;
45
46         return description;
47     }
48
49     public int compareTo(CD other)
```

```
50 {  
51     return this.artist.compareTo(other.artist);  
52 }  
53  
54 }  
55
```

```
1  /* Name: Jinku Jung
2   * UNI: dj2598
3   * Assignment: In-class Exercise #6
4   * Date: 4/8/2021
5   */
6
7  /**
8   // CDCollection.java    Author: Lewis/Loftus (modified by cannon)
9   //
10  // Represents a collection of compact discs.
11  /**
12
13  import java.text.NumberFormat;
14  import java.util.Locale;
15  import java.util.ArrayList;
16  import java.util.Collections;
17
18  public class CDCollection
19  {
20      private ArrayList<CD> collection;
21      private double totalCost;
22      private ArrayList<String> badArtists;
23      //-----
24      // Constructor: Creates an initially empty collection.
25      //-----
26      public CDCollection ()
27      {
28          collection = new ArrayList<CD>();
29          totalCost = 0.0;
30          badArtists = new ArrayList<String>();
31      }
32
33      //-----
34      // Adds a CD to the collection, as long as they aren't a bad artist.
35      // Else, an error message is thrown and the CD list stops being
36      // updated. However, an output of the user's CD Collection up to
37      // that errant Artist's CD is still outputted for the user to enjoy.
38      //-----
39      public void addCD (String title, String artist, double cost, int tracks)
40      throws BadMusicException {
41
42          for(String badArtist: badArtists)
43          {
44              if(artist.equals(badArtist))
45              {
46                  throw new BadMusicException("\nERROR: "
47                  + artist + " is crap!\n" +
48                  "\nYou can't possibly add this to your " +
49                  "CD collection ...\n")
```

```

50         + "\nPlease omit this artist and try updating your "
51         + "CD collection again.\n\nAny remaining "
52         + "*good* artists queued after " + artist
53         + " will not be added to your CD collection.\n"
54         + "\n--Your Computer\n");
55     }
56 }
57 collection.add( new CD (title, artist, cost, tracks));
58 totalCost += cost;
59 }
60
61 //-----
62 // Returns a report describing the CD collection.
63 //-----
64 public String toString()
65 {
66     NumberFormat fmt = NumberFormat.getCurrencyInstance(Locale.US);
67
68     String report = "~~~~~Wow Cool~~~~~\n";
69     report += "My CD Collection\n\n";
70
71     report += "Number of CDs: " + collection.size() + "\n";
72     report += "Total cost: " + fmt.format(totalCost) + "\n";
73     report += "Average cost: " + fmt.format(totalCost/collection.size());
74
75     report += "\n\nCD List:\n\n";
76
77     for (CD element : collection)
78         report += element.toString() + "\n";
79
80     return report;
81 }
82
83 public void sort()
84 {
85     Collections.sort(collection);
86 }
87
88 //-----
89 // Adds a bad artist to the bad artist blacklist, "badArtists."
90 //-----
91 public void addBad (String artist)
92 {
93
94     badArtists.add(artist);
95
96 }
97
98
99 }
100

```